

Important notes

- The computational processing must be made in the **R programming language**.
- The report must be written concisely in $\text{T}_{\text{E}}\text{X}$, **markdown** (R report format) or MS Word. It must contain the plots and the analysis based on the results.
- The report must be named as **FirstSurnameSecondSurname_hwk2_SMWR_2025-II.pdf** (MoralesMarin_hwk2_SMWR_II.pdf). This must be sent to lmoralesm@unal.edu.co with the subject **hwk2_SMWR**

Time series analysis

For the following hydrological time series:

- annual average streamflow,
- monthly average streamflow, and
- monthly precipitation,

whose length must be at least *20 years*, acquired from *IDEAM*, *CAR*, *IDIGER*, or other national or regional agencies, do the following:

1. Plot the *time series*, the *histogram*, and the *cumulative histogram*. Analyze and discuss the results.
2. Estimate the *mean*, *variance*, *skewness*, *kurtosis*, and *interquartile range*. Organize the results in tables and analyze them.
3. Determine the presence of a *trend* in the series. Check whether the trend is statistically significant using, for example, the *Mann–Kendall test*. Discuss the results.
4. Compute the *spectral density function* or *periodogram* of the series. Identify the main harmonics and fit a sinusoidal model if periodicity affects the stationarity of the series. Discuss the results.
5. If this is the case, remove the trend and/or periodicity from the time series and ensure that the series is *stationary*. Plot the treated series.
6. Compute and plot the *autocorrelation function (ACF)* and the *partial autocorrelation function (PACF)* with confidence limits. Analyze their shapes and make inferences about the possible model type and order.
7. Fit a *Box–Jenkins ARIMA* model to the residual time series. Report the model order and parameters, *AIC* and *BIC* criteria and discuss the results.
8. Plot *model residuals* and test residuals for *normality* and *randomness*. Discuss the results.
9. Reproduce the original series using the fitted model and plot the results.
10. Use the model to forecast the time series 10 units ahead and plot the results with the confidence intervals for 75% and 90%.